Hamza Mustafa

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Education

Skidmore College Saratoga Springs, NY

Bachelor of Arts in Computer Science, Minor in Mathematics, ***Major GPA: 3.95*** *Aug. 2023 – May 2027*

- **Coursework:** Data Structures, Algorithms, Software Design, Programming Languages, AI, Machine Learning

Software Engineering Experience

Backend Software Engineer July 2025 – Sept. 2025

ProgressSoft Corporation *Amman, Jordan*

- Cut average API latency by **25%** by using request-level timing **metrics** to find slow endpoints, then optimizing SQL queries and service logic in **Spring Boot** services backed by **PostgreSQL** and MySQL
- Secured **8+ REST API endpoints** for a financial transactions platform by implementing **JWT-based authentication** and role-based authorization, closing access gaps between internal and client-facing routes
- Raised service **code coverage** to **80%** by writing **JUnit** unit and integration tests and wiring them into the **CI/CD** pipeline as an **automated test suite** that ran on every merge request
- Built layered, **object-oriented** services (controller, service, repository) and containerized them with **Docker**, giving partner teams a consistent way to deploy and integrate their web-based tools

Projects

Shift2Sakura | *TypeScript, Next.js, Express.js, PostgreSQL, Prisma* Jan. 2026 – Apr. 2026

- Built a full-stack AI resume converter that turns English resumes into Japan-standard Rirekisho PDFs in **under 60 seconds** by chaining a local **LLaMA 3** model (Ollama) through an extract, translate, and format pipeline
- Used **Claude Code** and **GitHub Copilot** to scaffold API routes and Prisma migrations, then **reviewed and rewrote AI-generated code** that mishandled auth edge cases before merging
- Secured every API route with Clerk **JWT** middleware and designed a **PostgreSQL** schema with Prisma migrations and user-scoped queries so each user can only access their own conversion history

Checkers RL Agents | *Java, React* Jan. 2024 – May 2024

- Designed an **object-oriented** checkers engine in **Java** with separate board, move-generation, and agent classes behind a shared interface, so learning and search agents plug into the same game loop
- Trained a linear function approximation agent from scratch through self-play, then benchmarked it against an alpha-beta search agent over **1,000+ games** to compare win rates and decision behavior

Research Experience

Quantitative Researcher, Bloomberg Terminal Program May 2026 – Aug. 2026

Skidmore College *Saratoga Springs, NY*

- Trained PPO reinforcement learning agents in **Python** on SPY data that **outperformed buy-and-hold by 20%** in backtests, building a reproducible pipeline for data processing, training, and P&L evaluation
- Presented "Can AI Learn to Trade Options Without Seeing Options" to **120+ attendees**

Reinforcement Learning Research Assistant, NVIDIA Isaac Lab Sept. 2024 – Aug. 2025

Skidmore College *Saratoga Springs, NY*

- Improved gait stability under external forces by **40%** over baseline controllers by engineering reinforcement learning pipelines in **NVIDIA Isaac Lab** that pair learned policies with a PID controller
- Reduced training time by **30%** and improved convergence across **4,096 parallel environments** by redesigning reward functions and tuning simulation parameters and control policies
- **Co-authored** "Positive Rewards Alone Enable Bipedal Standing for Unitree Go2 in Isaac Lab," accepted at the LSU Symposium on Control, Learning, and Intelligent Systems

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL (PostgreSQL, MySQL), C, C#, HTML/CSS, R
Frameworks: Spring Boot, Next.js, React, Node.js, Express.js, REST APIs, Prisma ORM
Testing & DevOps: JUnit, unit & integration testing, automated test suites, Docker, CI/CD, Git, GitHub
AI & ML: PyTorch, NVIDIA Isaac Lab, Ollama (LLaMA 3), pandas, NumPy, Claude Code, GitHub Copilot
Practices: Object-oriented design, relational schema design, Agile/Scrum, code review, performance metrics